To Build Our Future, We Must Know Our Past: Contextualizing Paradigm Shifts in Natural Language Processing

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Abstract

NLP is in a period of disruptive change that is impacting our methodologies, funding sources, and public perception. In this work, we seek to understand how to shape our future by better understanding our past. We study factors that shape NLP as a field, including culture, incentives, and infrastructure by conducting long-form interviews with 26 NLP researchers of varying seniority, research area, institution, and social identity. Our interviewees identify cyclical patterns in the field, as well as new shifts without historical parallel, including changes in benchmark culture and software infrastructure. We complement this discussion with quantitative analysis of citation, authorship, and language use in the ACL Anthology over time. We conclude by discussing shared visions, concerns, and hopes for the future of NLP. We hope that this study of our field's past and present can prompt informed discussion of our community's implicit norms and more deliberate action to consciously shape the future.

1 Introduction

Natural language processing (NLP) is in a period of flux. The successes of deep neural networks and large language models (LLMs) in NLP coincides with a shift not only in the nature of our research questions and methodology, but also in the size and visibility of our field. Since the mid-2010s, the number of first-time authors publishing in the ACL Anthology has been increasing exponentially (Figure 1). Recent publicity around NLP technology, most notably ChatGPT, has brought our field into the public spotlight, with corresponding (over-)excitement and scrutiny.

In the 2022 NLP Community Metasurvey, many NLP practicioners expressed fears that private firms exert excessive influence on the field, that "a majority of the research being published in NLP is

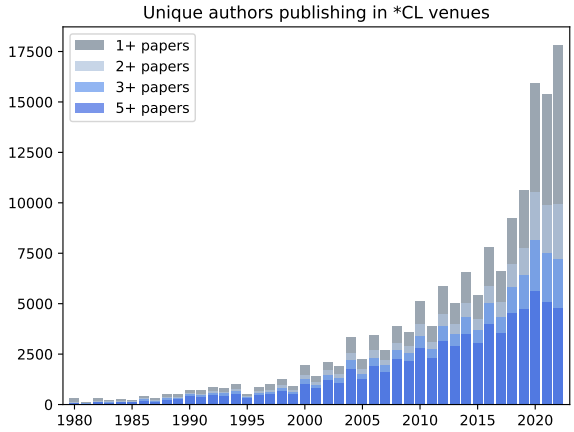


Figure 1: The number of unique researchers publishing in ACL venues has increased dramatically, from 715 unique authors in 1980 to 17,829 in 2022.

of dubious scientific value," and that AI technology could lead to a catastrophic event this century (Michael et al., 2022). More recently, there has been discussion of the increasing prevalence of closed-source models in NLP and how that will shape the field and its innovations (Rogers, 2023; Solaiman, 2023; Liao and Vaughan, 2023). In order to tackle these big challenges, we must understand the factors — norms, incentives, technology and culture — that led to our current crossroads.

We present a study of the community in its current state, informed by a series of long-form retrospective interviews with NLP researchers. Our interviewees identify patterns throughout the history of NLP, describing periods of research productivity and stagnation that recur over decades and appear at smaller scale around prominent methods (§3). Interviewees also point out unparalleled shifts in the community's norms and incentives. Aggregating trends across interviews, we identify key factors shaping these shifts, including the rise and persistence of benchmarking culture (§4) and the maturation and centralization of software infrastructure (§5). Our quantitative analysis of citation patterns, authorship, and language use in the ACL Anthology over time provides a complemen-

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tary view of the shifts described by interviewees, grounding their narratives and our interpretation in measurable trends. Through our characterization of the current state of the NLP research community and the factors that have led us here, we aim to offer a foundation for informed reflection on the future that we as a community might wish to see.

2 Methods

2.1 Qualitative methods

We recruited 26 researchers to participate in interviews using *purposive* (Campbell et al., 2020) and *snowball* sampling (Parker et al., 2019): asking participants to recommend other candidates, and purposively selecting for diversity in affiliation, career stage, geographic position, and research area (see participant demographics in Appendix A.1). Our sample had a 69-31% academia-industry split, was 19% women, and 27% of participants identified as part of a minoritized group in the field. Of our academic participants, we had a near-even split of early-, mid-, and late-career researchers; industry researchers were 75% individual contributors and 25% managers.

Interviews were semi-structured (Weiss, 1995), including a dedicated notetaker, and recorded with participant consent, lasting between 45-73 minutes (mean: 58 minutes). Interviews followed an interview guide (see Appendix A.3) and began by contrasting the participant's experience of the NLP community at the start of their career and the current moment, then moved to discussion of shifts in the community during their career. Interviews were conducted between November 2022 and May 2023; these interviews were coincidentally contemporaneous with the release of ChatGPT in November 2022 and GPT-4 in March 2023, which frequently provided points of reflection for our participants.

Following procedures of grounded theory (Strauss and Corbin, 1990), the authors present for the interview produced analytical memos for early interviews (Glaser et al., 2004). As interviews proceeded, authors began a process of independently *open coding* the data, an *interpretive* (Lincoln et al., 2011) analytical process where researchers assign conceptual labels to segments of data (Strauss and Corbin, 1990). After this, authors convened to discuss their open codes, systematizing recurring themes and contrasts to construct a preliminary closed coding frame (Miles and Huberman, 1994). After this, an author who was not present in the interview applied the closed coding frame to the data. In weekly analysis meetings, new codes arose to capture new themes or provide greater specificity, in which case the closed coding scheme was revised, categories refined, and data re-coded in an iterative process. Analysis reported here emerged first from this coded data, was refined by subsequent review of raw transcripts to check context, and developed in discussion between all authors.

2.2 Quantitative Methods

We use quantitative methods primarily as a coherence check on our qualitative results. While our work is largely concerned with the *causes* and *community reception* of changes in the community, our quantitative analyses provide evidence that these changes have occurred. This includes analyzing authorship shifts (Figures 1 and 4), citation patterns (Figure 2), terminology use (Figure 2,3) in the ACL anthology over time; for more details on reproducing this analysis, see Appendix B.

3 Exploit-explore cycles of work

Our participants described cyclical behavior in NLP research following methodological shifts every few years. Many participants referred to these methodological shifts as "paradigm shifts", with similar structure, which we characterize as *explore* and *exploit* phases.[1]

First wave: exploit. Participants suggested that after a key paper, a wave of work is published that demonstrates the utility of that method across varying tasks or benchmarks. Interviewees variously describe this stage as *"following the bandwagon" (17)*, *"land grab stuff" (9)*, or *"picking the low-hanging fruit" (8)*. A researcher with prior experience in computer vision drew parallels between the rise of BERT and the computer vision community after AlexNet was released, where *"it felt like every other paper was, 'I have fine tuned ImageNet trained CNN on some new dataset' " (17)*. However, participants identified benefits of this *"initial wave of showing things work" (17)* in demonstrating the value of techniques across tasks or domains; in finding the seemingly obvious ideas which do *not* work, thus exposing new areas to investigate; and in developing downstream applications. When one researcher was asked to

[1] Inspired by the verbiage of reinforcement learning, e.g. Sutton and Barto (2018).